

Set B

Question - 1

(a)

$$\Delta_{ADC} = \frac{V_{max} - V_{min}}{2^N}$$

$$= \frac{14 - 0}{2^2}$$

$$\Delta_{ADC} = 3.5V$$

when,

$$V_{in} = 0V$$

(reaches lower peak)

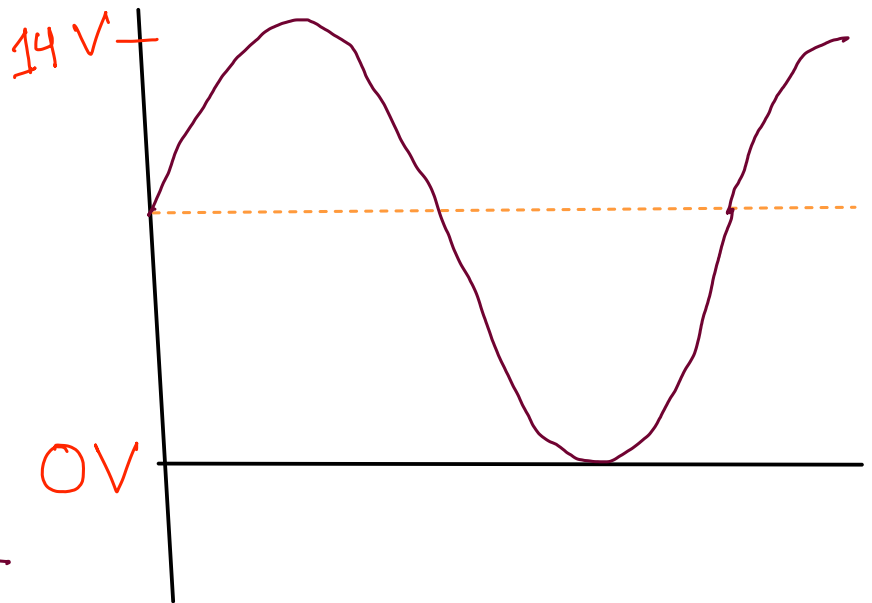
$$Y_1 = 0 ; Y_0 = 0$$

when,

$$V_{in} = 14V$$

(reaches higher peak)

$$Y_1 = 1 ; Y_0 = 1$$



Q. range	Q. level	encoded binary
0 - 3.5	1.75	00
3.5 - 7	5.25	01
7 - 10.5	8.75	10
10.5 - 14	12.25	11

Now,

$$V_{out} = -\frac{R_F \cdot V_{ref}}{R} \left(Y_1 + \frac{Y_0}{2} \right)$$

$$\therefore V_{00} = \frac{-5 \times (4V)}{5} \left(0 + \frac{0}{2} \right)$$
$$= 0V$$

$$\therefore V_{11} = \frac{-5 \times (4V)}{5} \left(1 + \frac{1}{2}\right)$$

$$= -6V$$

when $V_{in} = 0$	when $V_{in} = 14V$
$\% \text{ error} = \frac{0 - 0 }{0} \times 100\%$ $= \text{undefined}$	$\% \text{ error} = \frac{14 - -6 }{14} \times 100\%$ $= 57.14\%$

(b)

ADC step size	DAC step size
$\Delta_{ADC} = \frac{14 - 0}{2^2}$ $= 3.5V$	$V_{01} = -\frac{5 \times (4V)}{5} \left(0 + \frac{1}{2}\right)$ $= -2V$

Yes, they are different, Suggested steps to match them $\rightarrow$ we shall take magnitude value $|-3.5| = 3.5V$; $|-2| = 2V$

① V_{Ref} increase $\rightarrow$

$$V_{01} = \left| -\frac{5 \times V_{Ref}}{5} \left(0 + \frac{1}{2}\right) \right|$$

$$\Rightarrow 3.5 = V_{Ref} \times \frac{1}{2}$$

$$\therefore V_{Ref} = 7V \left[V_i = V_o = 7V \text{ for logic high now} \right]$$

② R_F increase $\rightarrow$

$$V_{01} = \left| -\frac{R_F' \times 4V}{5} \left(0 + \frac{1}{2}\right) \right|$$

$$\Rightarrow 3.5 = R_F' \times \frac{2}{5}$$

$$\therefore R_F' = 8.75k\Omega$$

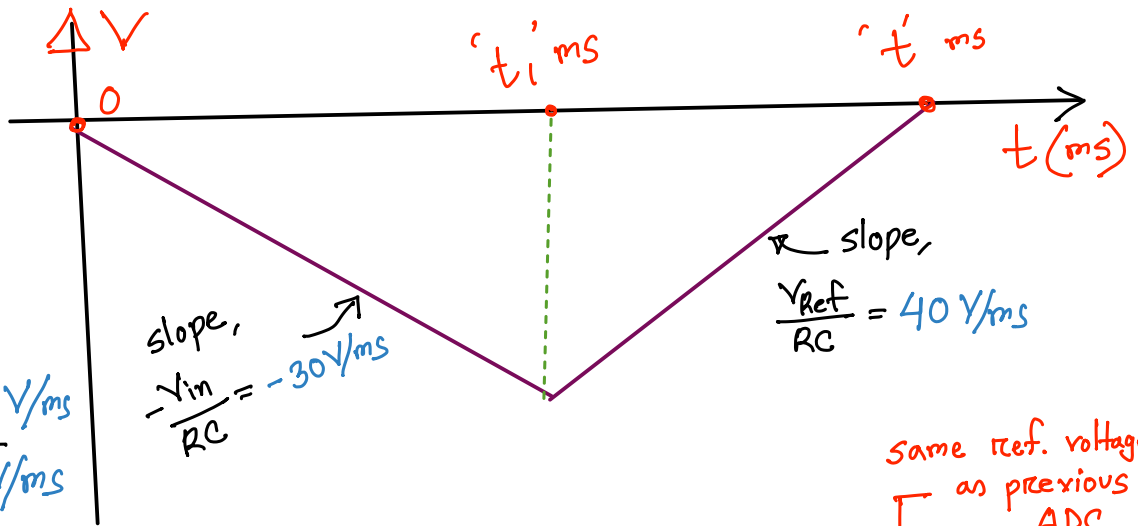
(c)

Thus,

$$\frac{-V_{in}}{RC}$$

$$= \frac{-30 \text{ V/ms}}{40 \text{ V/ms}}$$

$$\frac{V_{Ref}}{RC}$$



same ref. voltage
as previous
ADC

$$\Rightarrow \frac{V_{in}}{V_{Ref}} = \frac{3}{4} \quad \Rightarrow V_{in} = \frac{3}{4} \times V_{Ref} = \frac{3}{4} \times 14 \text{ V}$$

$\therefore V_{in} = 10.5 \text{ V}$ is the specific input,

we know, $V_{in} = m \times \text{step-size}$

$$\Rightarrow m = \frac{V_{in}}{\text{step-size}}$$

$$\Rightarrow m = \frac{10.5 \text{ V}}{\frac{14-0}{2^6}}$$

6-bit dual slope ADC

$$\therefore m = 48$$

$$(48)_{10} = (110000)_2$$

is the corresponding

digital o/p of the $V_{in} = 10.5 \text{ V}$

please
note

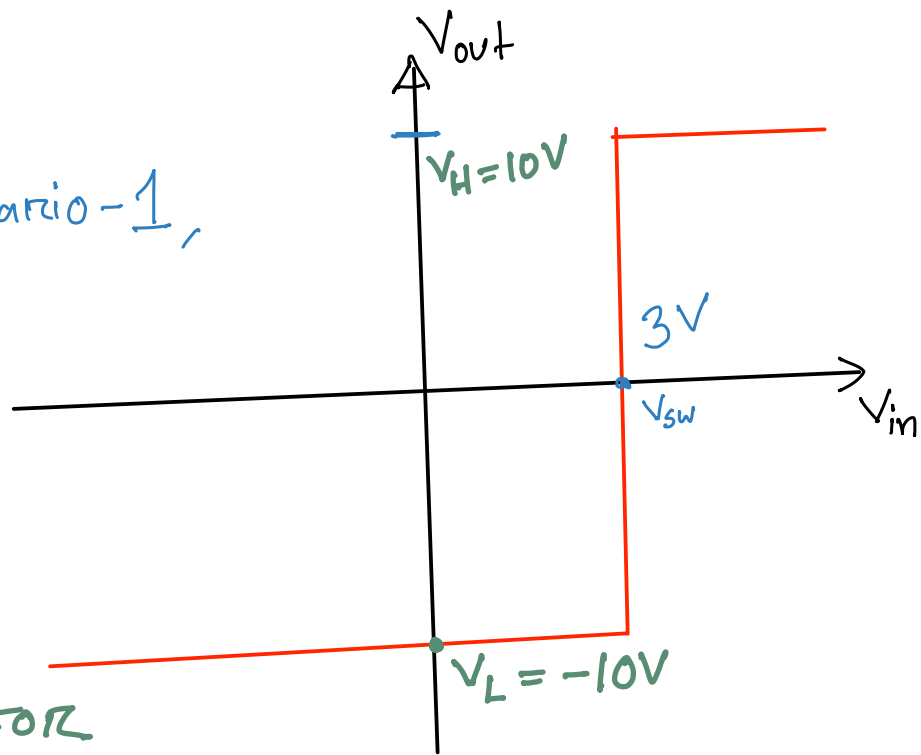
if your formula is, $V_{in} = (m+1) \times \text{step size}$

then, $m = 47$

$$\therefore m = 101111$$

Question-2

(a) As per the scenario-1,



This is the VTC of a Non-inverting comparator

$$\text{Thus, } -\frac{R_2}{R_1} \cdot V_{Ref} = V_{sw}$$

$$\Rightarrow -\frac{R_2}{R_1} (-5) = 3$$

$$\Rightarrow \frac{R_2}{R_1} = \frac{3}{5}$$

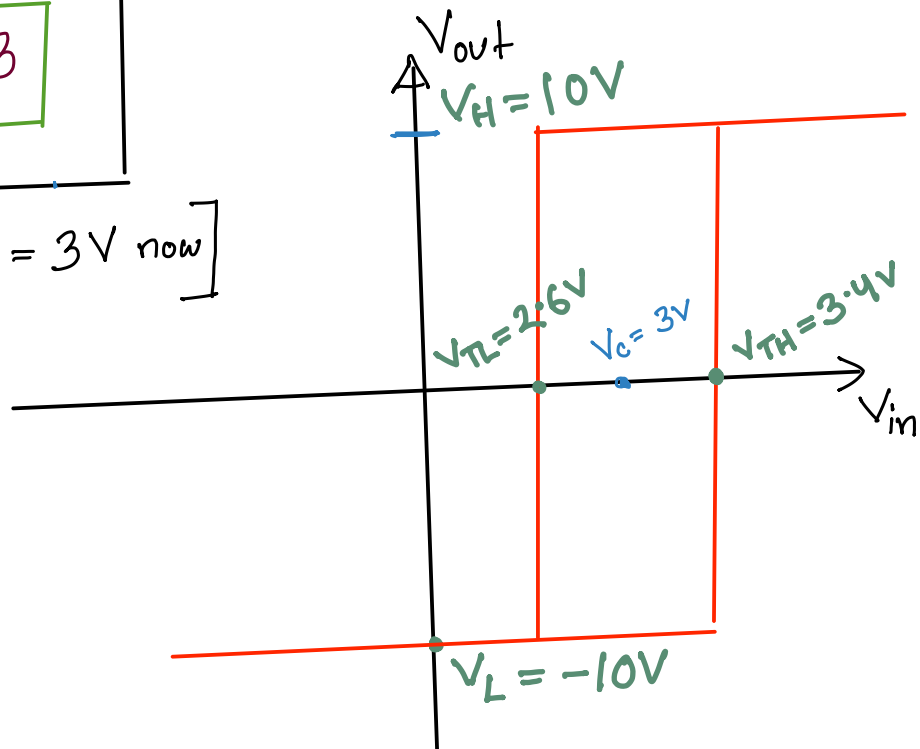
$$\Rightarrow \frac{R_1}{R_2} = \frac{5}{3}$$

$$\therefore R_1 : R_2 = 5 : 3$$

(b)

0.8V peak to peak noise consideration = hysteresis of 0.8V

So, at both sides of $V_{sw} = 3V$, there will be $\frac{0.8}{2} = 0.4$ volt of margin

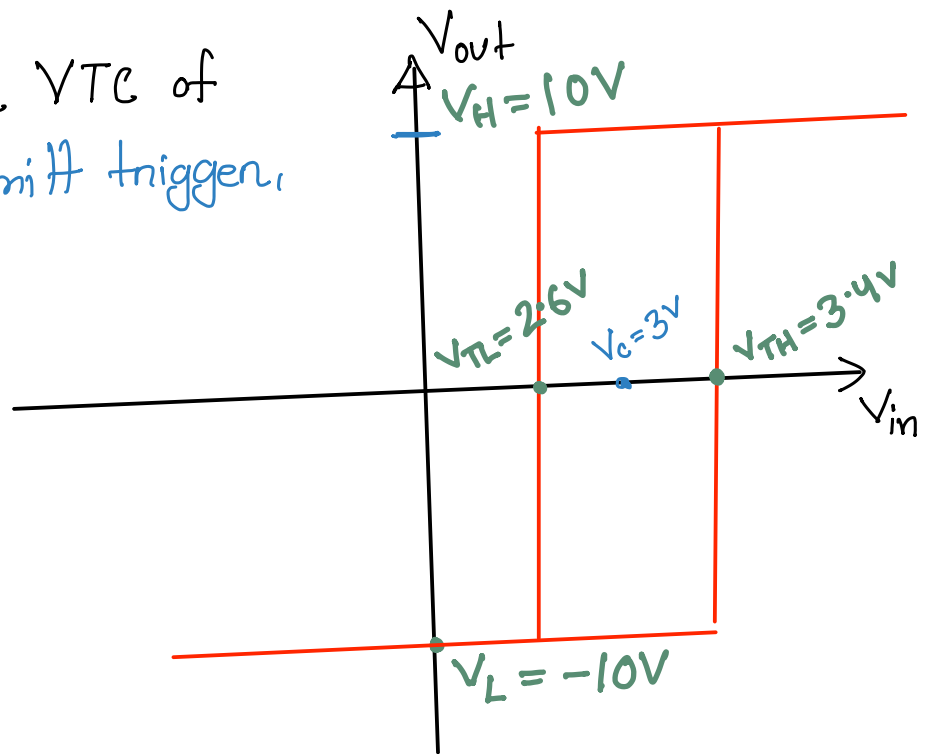


$$\therefore V_{TH} = 3 + 0.4 = 3.4 \text{ volt}$$

$$[V_c = 3V \text{ now}]$$

$$\therefore V_{TL} = 3 - 0.4 = 2.6 \text{ volt}$$

(c) This is now the VTC of a Non-inverting schmitt trigger, with Ref. voltage,



we know formulas,

$$V_{TH} = \frac{R_1 + R_2}{R_2} V_{Ref} - \frac{R_1}{R_2} V_L \quad \text{..... (i)}$$

$$V_{TL} = \frac{R_1 + R_2}{R_2} V_{Ref} - \frac{R_1}{R_2} V_H \quad \text{..... (ii)}$$

$$\begin{aligned} (V_{TH} - V_{TL}) &= -\frac{R_1}{R_2} V_L + \frac{R_1}{R_2} V_H \\ &= \frac{R_1}{R_2} (V_H - V_L) \end{aligned}$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{V_{TH} - V_{TL}}{V_H - V_L}$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{3.4 - 2.6}{10 - (-10)}$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{0.8}{20}$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{1}{25}$$

$$\therefore R_2 = 25 R_1$$

$$\text{let, } R_1 = 1k\Omega$$

$$\therefore R_2 = 25k\Omega$$

[(i) - (ii) is done]

stiff voltage

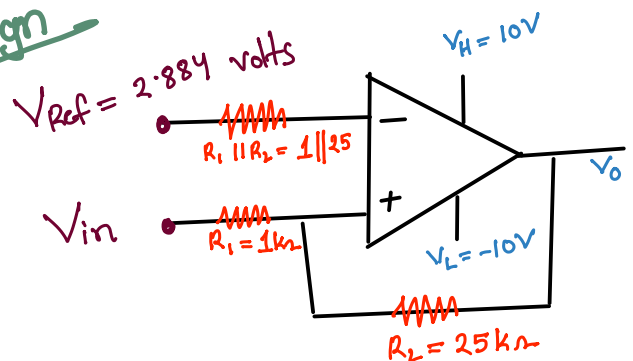
$$\text{Now, } \frac{R_1 + R_2}{R_2} V_{Ref} = \frac{3.4 + 2.6}{2}$$

$$\Rightarrow \frac{1 + 25}{25} \cdot V_{Ref} = 3$$

$$\Rightarrow V_{Ref} = \frac{3 \times 25}{26}$$

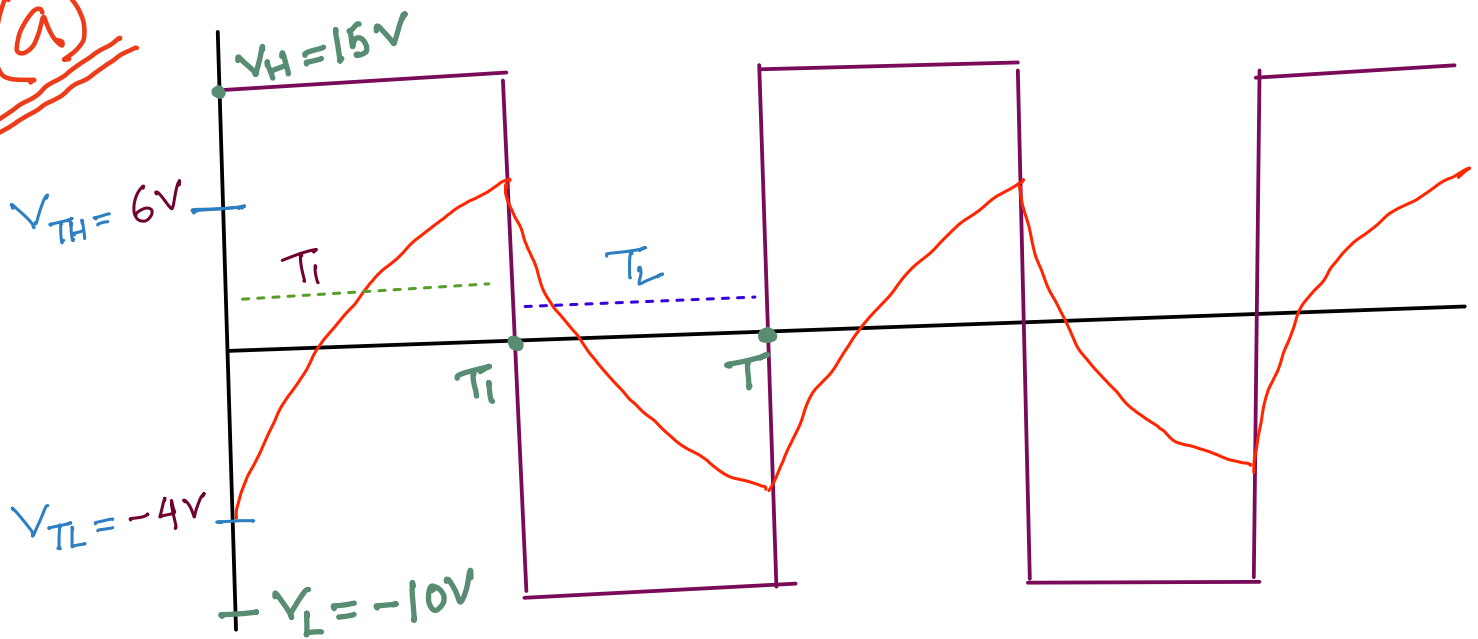
$$\therefore V_{Ref} = 2.884 \text{ volts}$$

The design



Question-3

(a)



given, $f = 600\text{Hz} \Rightarrow T = \frac{1}{f} = \frac{1}{800} = 1.67 \times 10^{-3} \text{sec}$
 $= 1.67 \text{ms}$

Now, $T = T_1 + T_2$

$$= \tau \ln \frac{V_H - V_{TL}}{V_H - V_{TH}} + \tau \ln \frac{V_L - V_{TH}}{V_L - V_{TL}}$$

$$\Rightarrow 1.67 \times 10^{-3} = \tau \left[\ln \frac{15 + 4}{15 - 6} + \ln \frac{-10 - 6}{-10 + 4} \right]$$

$$\Rightarrow 1.67 \times 10^{-3} = \tau \times 1.728$$

$$\therefore \tau = R_x \cdot C_x = 9.664 \times 10^{-4} \text{sec}$$

$$\begin{aligned}
 \text{Now, high time, } T_1 &= T \cdot \ln \frac{V_H - V_{TL}}{V_H - V_{TH}} \\
 &= 9.664 \times 10^{-4} \cdot \ln \frac{15+4}{15-6} \\
 &= 7.221 \times 10^{-4} \text{ sec} \\
 &= 0.722 \text{ ms}
 \end{aligned}$$

$$\begin{aligned}
 \text{Now, duty cycle, DC} &= \frac{T_1}{T} \times 100\% \\
 &= \frac{0.722}{1.67} \times 100\% \\
 &= \boxed{43.23\%}
 \end{aligned}$$

Now,

$$V_{TH} = \frac{R_1}{R_1 + R_2} V_H$$

$$\Rightarrow 6 = \frac{R_1}{R_1 + R_2} \cdot 15$$

$$\Rightarrow \frac{2}{5} = \frac{R_1}{R_1 + R_2}$$

$$\Rightarrow 2R_1 + 2R_2 = 5R_1$$

$$\Rightarrow \boxed{\frac{R_1}{R_2} = \frac{2}{3}}$$

$$V_{TL} = \frac{R_1}{R_1 + R_2} V_L$$

$$\Rightarrow -4 = \frac{R_1}{R_1 + R_2} \cdot (-10)$$

$$\Rightarrow \frac{2}{5} = \frac{R_1}{R_1 + R_2}$$

$$\Rightarrow \boxed{\frac{R_1}{R_2} = \frac{2}{3}}$$

(b) Now, given $f = 600 \times 2$
 $= 1200 \text{ Hz}$
 time period

V_H and V_L value not
 given for this
 only, $V_L = -V_H$ given

$$\therefore T = \frac{1}{f} = \frac{1}{1200} = 8.33 \times 10^{-4} \text{ sec}$$

$$= 0.833 \text{ ms}$$

Now, $V_{TH} = -\frac{R_1}{R_2} V_L$

$$\Rightarrow V_{TH} = -\frac{4}{12} (-V_H)$$

$$\therefore V_{TH} = V_H/3$$

$$V_{TL} = -\frac{R_1}{R_2} V_H$$

$$\Rightarrow V_{TL} = -\frac{4}{12} V_H$$

$$\therefore V_{TL} = -V_H/3$$

Now,

$$T = T_1 + T_2$$

$$\Rightarrow T = \tau \left(\frac{V_{TH} - V_{TL}}{V_H} \right) + \tau \left(\frac{V_{TL} - V_{TH}}{V_L} \right)$$

$$\Rightarrow T = \tau \left(\frac{V_H/3 + V_H/3}{V_H} \right) + \tau \left(\frac{-V_H/3 - V_H/3}{-V_H} \right)$$

$$\Rightarrow 8.33 \times 10^{-4} \text{ sec} = \left(\frac{2}{3} \tau + \frac{2}{3} \tau \right) = \frac{4}{3} \tau$$

$$\Rightarrow \tau = R_i \cdot C_i = 6.247 \times 10^{-4} \text{ sec}$$

$$\therefore C_i = \frac{6.247 \times 10^{-4} \text{ sec}}{30 \times 10^3 \Omega} = 2.082 \times 10^{-8} \text{ F} = 20.82 \text{ nF}$$

R_i value